Alteryx Workflow Document

Day04.yxmd

Report Create Date : 2026-04-25

Workflow Basic Information

FileName : Day04.yxmd
Workflow name : Day04
yxmd version : 2025.1
RunE2 : T
Workflow description :

Author Information

Author :
Company :
Copyright :

Workflow Meta information

Setting Name	Value
Root Tool Name	
Tool Version	
Tool In DB	
Tool In DB value	False
Category Name	
Search Tags	
Description Link actual	
Description Link displayed	
Workflow ID	2e4c80c0-c741-4623-a4a6-8c71d51ab24e
Platform Workflow ID	
Example Description	
Example File	
Telemetry Previous Workflow ID	1234fa68-e137-40e4-b530-8eb0eb668b3f
Telemetry Origin Workflow ID	f8905802-0a79-4237-ba5c-b8dfb6a1aa79

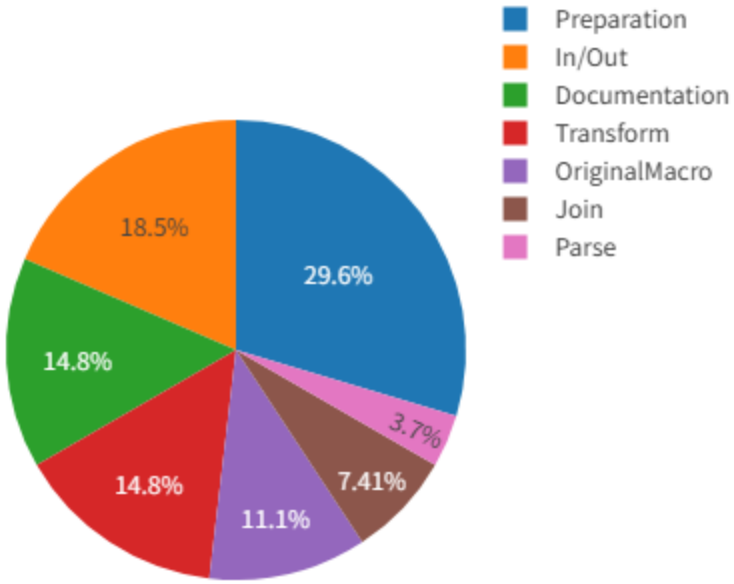
workflow detail setting

Setting Name	Value
yxmdVer	2025.1
RunE2(AMP Engine)	T
Memory_default	True
Global Record Limit	0
Temp Files	True
Annotation on	True
Annotation include Tool Name	False
Convert Error Limit	10
Convert Error Limit Stop	False
Cancel On Error	False
Disable Browse tool	False
Enable Performance Profiling	False
Run With E2	True
Default TZ	Local
Predictive Tools Code Page	932
Disable All Output	False
Show All Macro Messages	False
Show Connection Status Is On	True
Show Connection Status Only When Runninng	True
Zoom Level	0
Layout Type	Horizontal
Is Template	False
Name Is File Name	True

Tool statistics

Total Tool Count : 27
Number of tool types : 16

Category



Tool Category	Count
Preparation	8
In/Out	5
Documentation	4
Transform	4
OriginalMacro	3
Join	2
Parse	1

Tools

Tool Category	Tool Name	Count
Documentation	Comment	4
In/Out	Browse	3
Preparation	Filter	3
Preparation	Record ID	3
In/Out	Text Input	2
Transform	Cross Tab	2
Join	Append Fields	1
Join	Join	1
OriginalMacro	CheckOnMap	1
OriginalMacro	CreatePoints	1
OriginalMacro	aoc_202504_ite	1
Parse	RegEx	1
Preparation	Data Cleanse Pro	1
Preparation	Formula	1
Transform	Count Records	1
Transform	Summarize	1

Input/Output

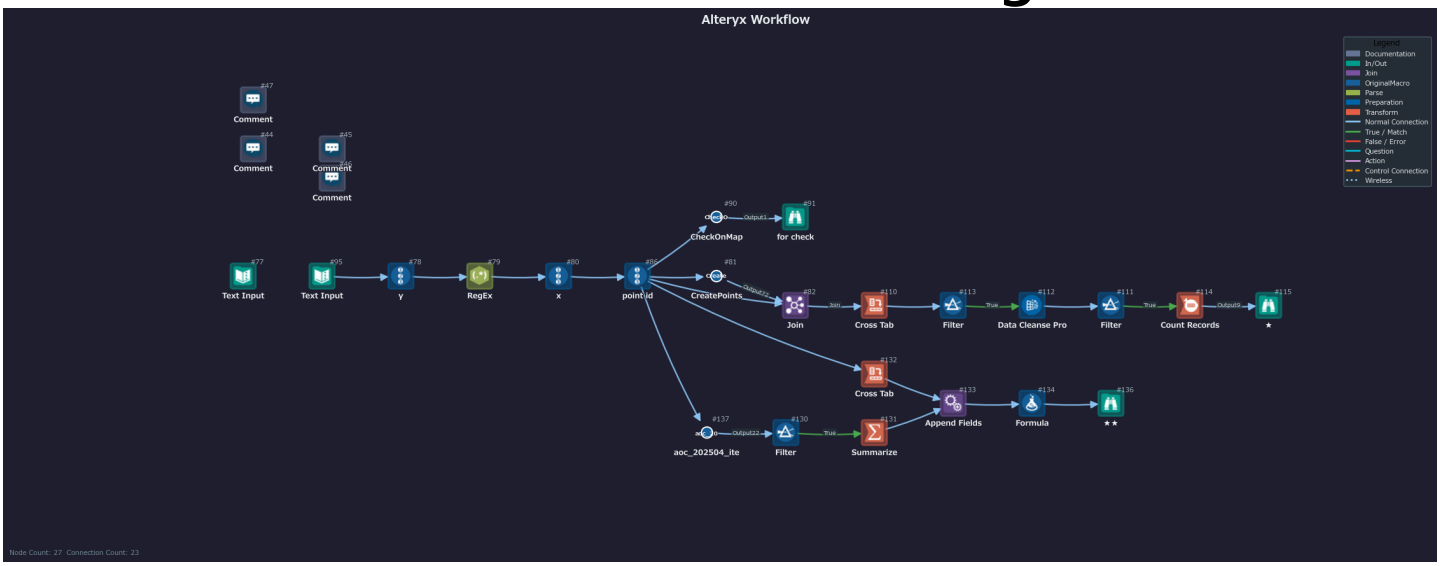
Input overview

Tool Name	ToolID	Input Type	Input
Text Input	77	Text	Text
Text Input	95	Text	Text

Output overview

Tool Name	ToolID	Output Type	Output
Browse	91	Browse	
Browse	115	Browse	
Browse	136	Browse	

Workflow Canvas Image



Alteryx Workflow detail description

Workflow Overview

This Alteryx workflow processes a grid of characters, represented as strings of '.' and '@', provided in a text input. It transforms this raw input into a structured dataset with x,y coordinates and unique identifiers for each character. The workflow then branches into two main analytical paths. The first path uses a custom macro to generate potential related points, joins this information back, and then performs a cross-tabulation to count neighboring '@' characters. It filters these counts to identify specific '@' cells based on a condition and provides a final count. The second path applies another custom macro for more complex processing, filters characters, and calculates a final difference based on overall and specific character counts.

Input/Output Data Overview

Input Data

The primary input data for the workflow comes from Text Input(95), which contains 135 rows of string data. Each string is composed of '.' and '@' characters, representing a large grid. An additional Text Input(77) is present with 10 rows of similar character data, but it is not connected to the main processing stream.

Output Data

The workflow produces several outputs for review and final results:
- Browse(91): Displays an intermediate view of the processed grid, showing character values pivoted into columns based on their x-coordinates for each y-coordinate row, originating from a macro named 'CheckOnMap'.
- Browse(115): Shows a single numerical value representing the count of specific '@' characters that meet a particular condition (their count of surrounding '@' characters is 3 or less). This is marked as '★'.
- Browse(136): Displays a single numerical value, which is the difference between the total count of '@' characters in the entire grid and a count of specific '@' characters identified through an advanced macro process. This is marked as '★★'.

Workflow Details

Section 1: Data Grid Transformation

This initial section takes the raw string input, assigns row (y) and column (x) identifiers, and then splits each string into individual characters to create a structured dataset where each character occupies its own record with associated grid coordinates and a unique identifier.

Tool Name	Tool ID	Processing Details
Text Input	95	Reads 135 rows of string data, each representing a line of characters ('.' or '@'), into a field named 'Input'.
Record ID	78	Adds a sequential integer identifier, named 'y', to each incoming record, starting from 1. This represents the row number.
RegEx	79	Splits the 'Input' string field from each record into individual characters, creating a new record for each character. This effectively transposes the grid data into a long format, with each character getting its own record.
Record ID	80	Assigns a sequential integer identifier, named 'x', to each record within groups defined by the 'y' field. This represents the column number for each character within its original row.

Tool Name	Tool ID	Processing Details
Record ID	86	Assigns a unique sequential integer identifier, named 'ID', to each record after the x,y coordinates have been established. This creates a unique ID for every single character cell in the grid.

Section 2: Generating and Joining Related Points

This section utilizes a custom macro to generate additional data points, likely representing neighboring cells or performing some spatial calculation based on the grid coordinates. The generated points are then recombined with the original grid data.

Tool Name	Tool ID	Processing Details
Macro	81	Executes the 'AoC¥CreatePoints.yxmc' macro. This macro likely generates new data points (e.g., neighbors) based on the input 'x', 'y', and 'Input' (character value) fields, outputting fields like 'x_1' and 'y_1'.
Join	82	Combines the data from the 'CreatePoints' macro (left input) with the main data stream (right input). Records are matched when the 'x_1' field from the left input equals the 'x' field from the right input, and 'y_1' equals 'y'. This associates the generated points with their corresponding grid cell data.
Macro	90	Executes the 'AoC¥CheckOnMap.yxmc' macro. This macro takes the x, y coordinates and the character value as input. It likely performs a spatial transformation or check, possibly converting the grid data into a wide format where columns represent x-coordinates for each y-coordinate. Its output is then sent to a browse tool for inspection.
Browse	91	Displays the data output from the 'CheckOnMap' macro for review, showing a cross-tabulated representation of the grid.

Section 3: Grid Analysis and Filtering (Part 1)

This section analyzes the character data, counting occurrences of '.' and '@' within specific contexts, and then filters the records based on a condition related to the count of '@' characters, culminating in a final count of matching records.

Tool Name	Tool ID	Processing Details
Cross Tab	110	Pivots the data, grouping records by 'ID', 'x', 'y', and the character 'value'. It creates new columns for each distinct 'Input' character ('.' and '@') and counts the occurrences of 'ID' for each. This provides a count of '.' and '@' for every unique grid cell, resulting in fields like 'CountWithNulls_.' and 'CountWithNulls_@'.
Filter	113	Filters the data, retaining only those records where the 'value' field (representing the character) is equal to '@'.
Data Cleanse	112	Cleans the numeric fields 'CountWithNulls_.' and 'CountWithNulls_@' by replacing any null values with zero. This ensures consistent numeric calculations downstream.
Filter	111	Filters the data further, keeping only records where the count of '@' characters ('CountWithNulls_@') is less than or equal to 3.
Macro	114	Executes the 'CountRecords.yxmc' macro. This utility macro counts the total number of records that satisfy the preceding filter conditions.
Browse	115	Displays the final count of records that passed through the filters, representing specific '@' characters meeting the criteria. This is marked as '★'.

Section 4: Grid Analysis and Calculation (Part 2)

This section involves another complex macro process to analyze the character grid, identifies specific '@' characters based on its internal logic, and then computes a final difference by subtracting this specific count from the overall count of '@' characters in the entire grid.

Tool Name	Tool ID	Processing Details
Macro	137	Executes the 'aoc_202504_ite.yxmc' macro. This macro likely performs an iterative process or a specific set of rules (possibly related to Advent of Code Day 4) on the grid data (ID, x, y, character value). It produces an output stream, which is then filtered.

Tool Name	Tool ID	Processing Details
Filter	130	Filters the data output from the 'aoc_202504_ite' macro, keeping only records where the 'Input' field (representing the character) is equal to '@'.
Summarize	131	Counts the number of 'ID' values for the filtered records, effectively providing a total count of specific '@' characters identified by the macro and subsequent filter.
Cross Tab	132	Calculates overall counts for '.' and '@' characters across the entire grid. It creates new columns for each distinct 'Input' character and counts the occurrences of 'ID' without any grouping, resulting in overall 'CountWithNulls_.' and 'CountWithNulls_@'.
Append Fields	133	Combines the single-record output from Summarize(131) (the count of specific '@' characters) with the single-record output from Cross Tab(132) (the overall counts of '.' and '@').
Formula	134	Calculates a new field named 'result' by subtracting the specific count of '@' characters (from Summarize) from the overall count of '@' characters (from Cross Tab).
Browse	136	Displays the final calculated 'result' value, which represents the difference between the two counts of '@' characters. This is marked as '★★'.